

Coding & Solution

Date	1 July 2026
Team ID	XXXXXX
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/nithyasowreddy/OptiCrop-Smart-Agricultural-Production-Optimization-Engine
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn
Key Features Implemented	AI-based crop recommendation, soil & weather analysis, responsive web interface, real-time prediction
Pending / Incomplete Features	Cloud deployment (optional), user authentication (future enhancement)
Setup / Run Instructions	Install dependencies using requirements.txt, then run python app.py and open the application in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The application is fully functional and successfully predicts suitable crops based on soil nutrients and environmental parameters. The code is organized, readable, reusable, and follows good development practices.